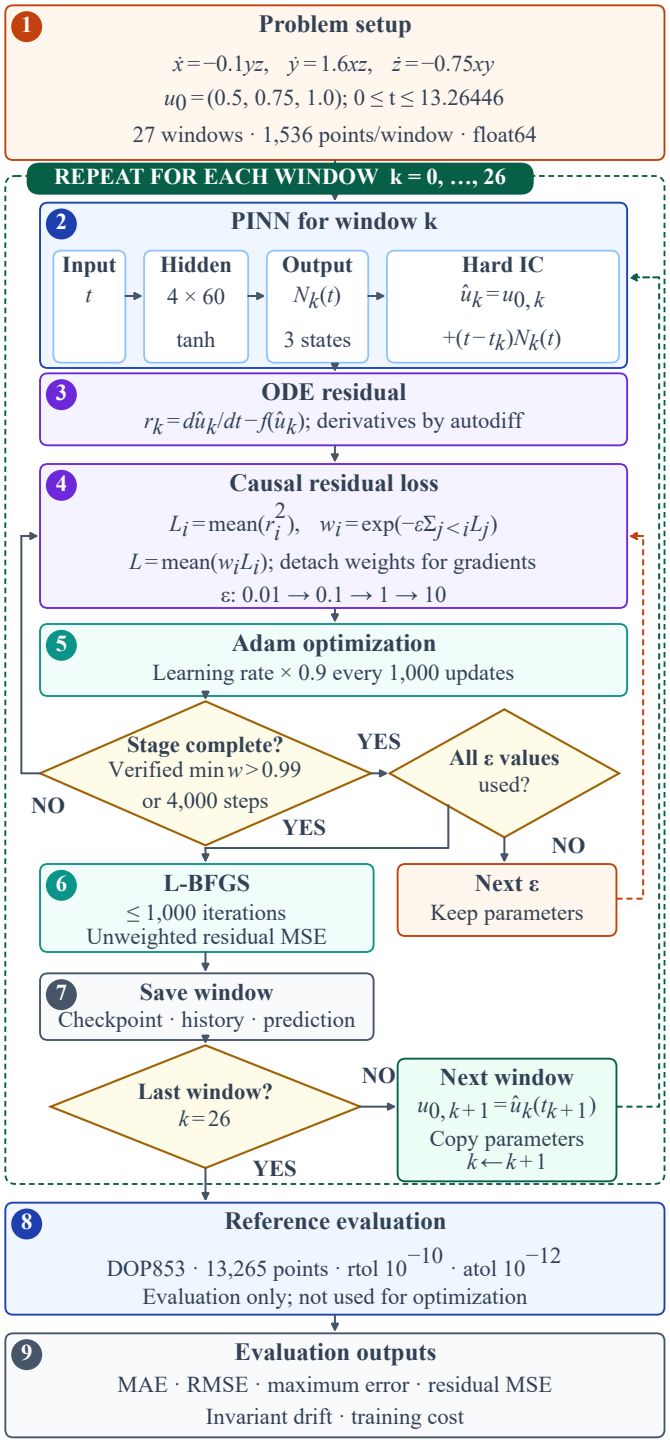

1
Problem setup
$\dot{x} = -0.1yz, \quad \dot{y} = 1.6xz, \quad \dot{z} = -0.75xy$
$u_0 = (0.5, 0.75, 1.0); \; 0 \le t \le 13.26446$
27 windows · 1,536 points/window · float64
REPEAT FOR EACH WINDOW k = 0, …, 26
2
PINN for window k
Input
t
Hidden
4 × 60
tanh
Output
$N_k(t)$
3 states
Hard IC
$\hat{u}_k = u_{0,k}$
$+(t - t_k)N_k(t)$
3
ODE residual
$r_k = d\hat{u}_k/dt - f(\hat{u}_k)$; derivatives by autodiff
4
Causal residual loss
$L_i = \mathrm{mean}(r_i^2), \quad w_i = \exp(-\varepsilon \Sigma_{j<i} L_j)$
$L = \mathrm{mean}(w_i L_i)$; detach weights for gradients
ε: 0.01 → 0.1 → 1 → 10
5
Adam optimization
Learning rate × 0.9 every 1,000 updates
Stage complete?
Verified min $w > 0.99$
or 4,000 steps
YES
NO
All ε values
used?
YES
NO
6
L-BFGS
≤ 1,000 iterations
Unweighted residual MSE
Next ε
Keep parameters
7
Save window
Checkpoint · history · prediction
Last window?
$k = 26$
NO
YES
Next window
$u_{0,k+1} = \hat{u}_k(t_{k+1})$
Copy parameters
$k \leftarrow k+1$
8
Reference evaluation
DOP853 · 13,265 points · rtol 10^{-10} · atol 10^{-12}
Evaluation only; not used for optimization
9
Evaluation outputs
MAE · RMSE · maximum error · residual MSE
Invariant drift · training cost